Year 2 Assessed Problems

Semester 2

Assessed Problems 1

SOLUTIONS TO BE SUBMITTED
ON CANVAS BY

Wednesday 4th February at 17:00

Assessed Problems 1

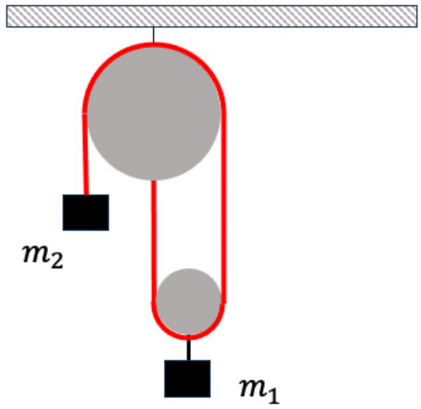
Richard Mason

Problem 1. Given the following set of simultaneous linear equations

$$\begin{aligned} x + 2y + 2z &= 3 \\ x + y &= 0 \\ 2x + y + 2z &= -5 \end{aligned}$$

1. Represent this system in the form $A\boldsymbol{x} = \boldsymbol{b}$, defining the matrix A, and the vectors $\boldsymbol{x}$ and $\boldsymbol{b}$ explicitly. **[3]**
2. Solve the set of equations via Gaussian elimination. Note: you should form the augmented matrix $(A, \boldsymbol{b})$. **[5]**
3. Consider modifying the fist equation to $tx + 2y + 2z = 3$. Is there a value of t for which these equations do not solve? **Hint**: try adding or subtracting pairs of equations to generate an inconsistent result. **[2]**

The diagram shows two masses, m_1 and m_2 attached to a pulley system shown. There is no friction in the system, all the ropes are massless and slide freely on all the pulley wheels shown. For convenience you can ignore the finite size of the pulley wheels.



(a) How many degrees of freedom does the system have? Give careful reasons for your answer. **[2]**

(b) Write down expressions for the kinetic and potential energy of the system and hence write down a Lagrangian for the system. **[3]**

(c) Find Lagrange's Equation(s) for the system. **[2]**

(d) Find the downward acceleration of the mass m_1. **[2]**

(e) Find the condition on the masses that the mass m_1 accelerates downwards. What happens if $m_1 = 2m_2$? **[1]**

Structure in the Universe (03 00554)

Problem Sheet 1

a) Derive an expression for the time taken, t_c for a spherical body of radius *r* to collapse, under the assumption that the supporting pressure is 'switched off' so that it goes into 'free fall'. Do so for the approximation where the acceleration during collapse is taken to be constant. (Hint: use a simple equation of motion to relate the t_c to *r* and *g* for any body in motion; then use an expression for *g*, which we assume remains constant, for a spherically symmetric body just before collapse and equate the two equations.) How will the acceleration vary in reality? **[5]**

b) A full treatment gives rise to the expression:

$$t_c = \frac{\pi}{\sqrt{8GM}} r^{3/2}.$$

By what fraction does your result differ from the accurate formula above? **[1]**

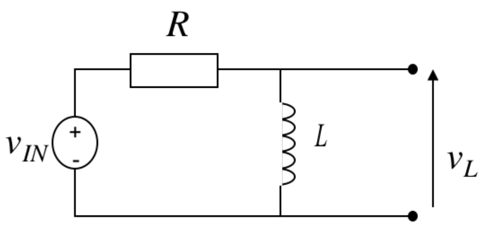
c) By considering the manner in which the collapse time varies for different shells, discuss how the density profile will change during the free collapse of:

1. A uniform density sphere, i.e., one for which $\rho(r)$ is constant at all *r*.
2. A sphere with a density profile $\rho(r) \propto r^{-1}$.

[Hint: first, re-express t_c in terms of $\rho(r)$.] **[4]**

STC filters

1. Consider the circuit below, with $L = 100\,\text{mH}$ and $R = 100\,\Omega$.



   (a) Test the circuit at $\omega = 0$ and $\omega = \infty$ and state what filter type it is. **[1]**

   (b) Derive the corresponding transfer function and write it in the general form. **[2]**

   (c) Sketch the Bode Plots for magnitude and phase for ω between $1\,\text{Hz}$ and $10^5\,\text{Hz}$. **[4]**

   (d) Consider a pulse waveform at the input with amplitude A and width T. Sketch the output as a function of time assuming $T >> L/R$. **[1]**

   (e) Using the formula for the transient response given in lecture 2, write the functional form of the output voltage for $0 \leq t < T$ and $t \geq T$. You can assume the case where $T >> L/R$. **[2]**